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جنة سيكتوم

Electrical Circuits

electron $\rightarrow m_e = 9.1 \times 10^{-31}$ kg

current $\rightarrow \boxed{i = \frac{dq}{dt}}$ A → derive with the respect of time → rate

↳ Orderly movement of free electrons

• Prefixes :-

10^{-3}	10^{-6}	10^{-9}	1	10^{3}	10^{6}	10^{9}
mA	μA	nA	A	kA	MA	GA

Current ∝ Direction

+ 5A → − (Conventional current)

+ ←5A− (5A > −5A)

Voltage (Potential difference) ∝ Polarity

voltage drop / voltage rise

+ → − → voltage drop

− → + → voltage rise

A is higher than B

$V = \frac{dw}{dq}$ "V"

$i = \frac{dq}{dt}$

$\int dq = \int i\,dt \Rightarrow q = \int_{t_1}^{t_2} i\,dt$

Q. Determine the total charge transfered over the time interval of $0 \le t \le 10$ s. When $i(t) = \frac{1}{2} t$ mA

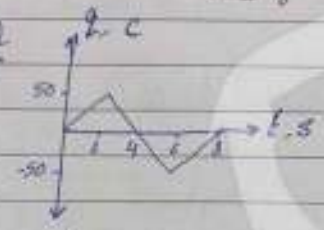
Solution:

$$i = \frac{dq}{dt} \rightarrow q = \int_0^{10} i\,dt$$

$$= \int_0^{10} \frac{1}{2} t\,dt$$

$$= \frac{1}{2} \frac{t^2}{2} \Big|_0^{10} = 25 \text{ mC}$$

Q.



The charge following in a wire is plotted below. Sketch the corresponding Current

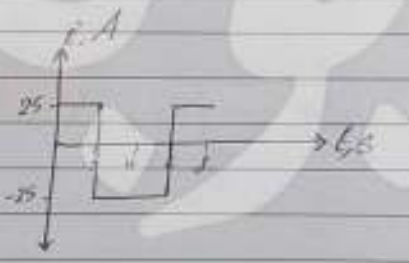
$$i = \frac{dq}{dt}$$

$(0-1)s \Rightarrow i = \frac{50-0}{2-0} = 25A$

$(2-4)s \Rightarrow i = \frac{0-50}{4-2} = -25A$

$(4-6)s \Rightarrow i = \frac{-50}{2} = -25A$

$(6-8)s \Rightarrow i = 25A$



Q



Find q at $1 \le t \le 2$.

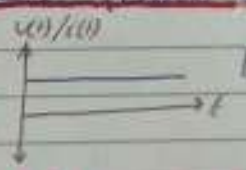
Solution:

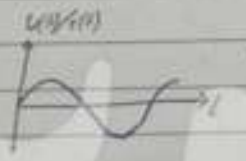
$q = \int i\,dt$ = area under curve ⇒



$q = A_1 + A_2 \Rightarrow q = (5 \times 1 \times \frac{1}{2}) + (5 \times 1) = 7.5\,C$

- **DC (Direct Current):-**





AC (Alternating Current)

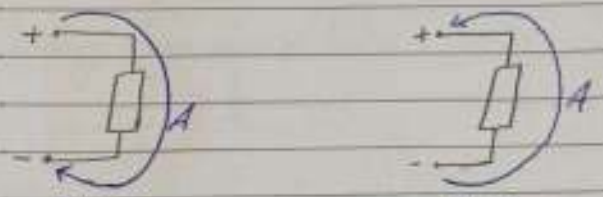
Power & Energy

W — J

$P = \frac{dw}{dt} \Rightarrow w = \int p\,dt$

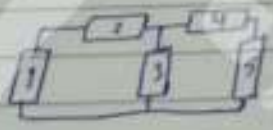
$\boxed{P = VI} \Rightarrow p = \frac{dw}{dq} \times \frac{dq}{dt}$

• Passive Convention Sign:-



To deliver
To supply
To release
To consume
To absorb

Q For the following circuit if $P_1 = -20w$, $P_2 = 60w$, $P_4 = 45w$, $P_5 = 30w$, Calculate the power P_3 received or delivered by element 3.



solution:

$\Sigma P = 0$

$-20 + 60 + 45 + 30 + P_3 = 0 \Rightarrow P_3 = 70w$ received by 3

Circuit Elements

battery Generator & Active

Passive

Voltage Source

Current Source

Independent

Dependent

Independent

Dependent

VCVS

CCCS

Basic Laws

Ohms Law: $V = iR$

Resistance $R(\Omega)$

Resistor

$$i = \frac{V}{R} = \frac{1}{R} V$$

$\frac{1}{R} = G$: Conductance "S"

$$i = GV$$

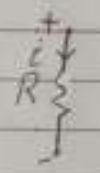
$V = iR$

$R = 0 \rightarrow V = 0$: Short Circuit: ——

i ?!

$R \rightarrow \infty \longrightarrow i = \frac{V}{R} = \frac{V}{\infty} = 0$: Open Circuit — o c —

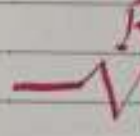
V ?!



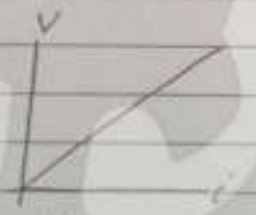
$V = iR$



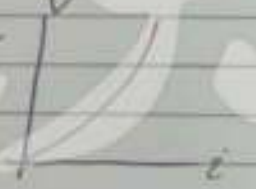
$V = -iR$



(linear resistor)



non linear resistor



$V = iR$

$P = Vi$
$= i^2R$
$= \frac{V^2}{R}$

$P > 0$

Two-terminal device



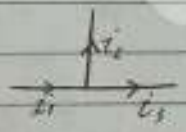
} Branch



} node



$\sum_{node} i = 0$

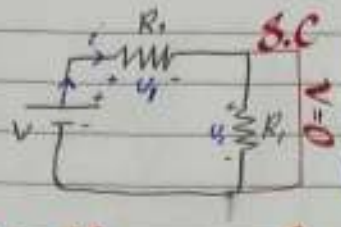


$i_1 = i_2 + i_3$

$\sum_{loop} V = 0$

} rise sign (-) } drop sign (+)

Series Circuit



$V_1 = iR_1$

$V_2 = iR_2$

$V_1 + V_2 = V$



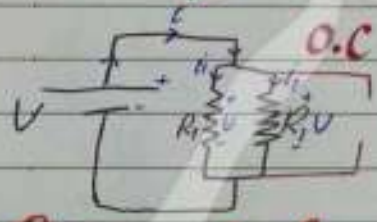
Voltage division Law

$$V_1 = \frac{VR_1}{R_1+R_2} \qquad V_2 = \frac{VR_2}{R_1+R_2}$$

$$R_{tot} = R_{eq} = \sum_{i=1}^{n} R_i = R_1 + R_2 + \dots + R_n$$



Parallel Circuit



Current division Law

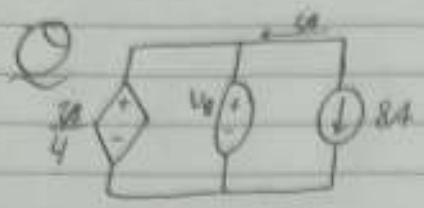
$$i_1 = \frac{i\,R_2}{R_1+R_2} \qquad i_2 = \frac{iR_1}{R_1+R_2}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

$$R_{eq} = \frac{R_1 R_2}{R_1+R_2}$$



Q

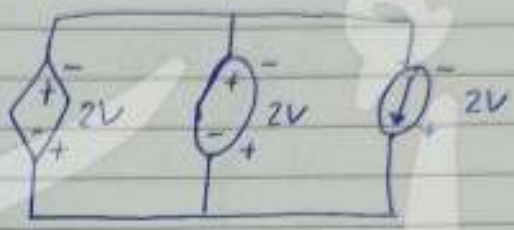


Find v_8 and P_{8A}.

Solution

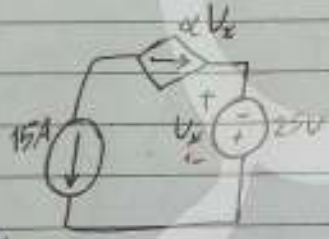
$i_8 = -8A$

$\frac{v_8}{4} = \frac{-8}{4} = -2V$



$P_{8A} = vi = (-2)(8) = -16$ watt

Q



Find α and P_{25V}.

Solution:

$v_x = -25$ V ⟶



$-v_x - 25 = 0 \Rightarrow v_x = -25$

$\alpha v_x = -15$

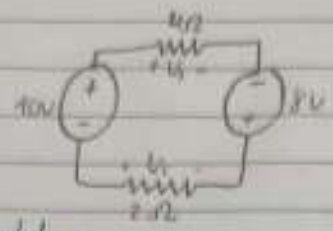
$\alpha(-25) = -15$

$\alpha = \frac{15}{25}$

$P_{25V} = v_i$

$= (25)(15) = 375$ watt

Q



Find V_1 and V_2.

Solution

$\Sigma V = 0$

$-10 + V_1 - 8 - V_2 = 0$

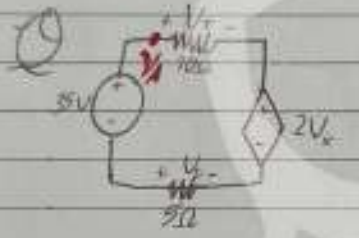
$-10 + 4i - 8 + 2i = 0$

$6i = 18$

$i = \frac{18}{6} = 3A$

$V_1 = 12v$, $V_2 = -6v$ ← قانون أوم

Q



Find V_x and V_0.

Solution:

$-35 + 10i + 2V_x + 5i = 0$

$V_x = 10i$

$-35 + 10i + 20i + 5i = 0 \Rightarrow i = 1A$

$V_x = 10v$, $V_0 = -5V$

$V_A = 35v$

Q



Find i_o and V_o.

solution:

$6 = i_o + i_x$ ... ①

$i_x = \frac{v_o}{4} + i_y$ ... ②

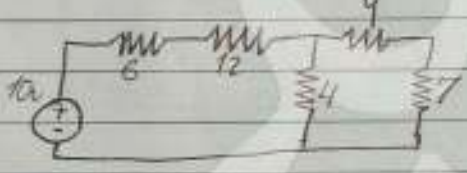
$V_o = 2i_o \quad | \quad i_y = \frac{V_o}{8}$

$6 = i_o + \frac{i_o}{4} + i_y$

$6 = i_o + \frac{i_o}{4} + \frac{2i_o}{8} \Rightarrow 6 = \frac{6i_o}{4} \Rightarrow i_o = 4A$

$V_o = 2 \cdot 4 = 8\ V$

Q



Find Req?

solution:

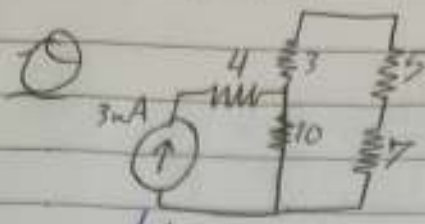
9 is in series with 7 → $9+7 = 16\Omega$

16 is parralled to 4 → $\frac{16 \cdot 4}{20} = \frac{16}{5}\ \Omega$

6 is in series with 12 and $\frac{16}{5}$ → $\frac{16}{5} + 12 + 6 = 21.2\ \Omega$



Q



Find Req.

Solution:

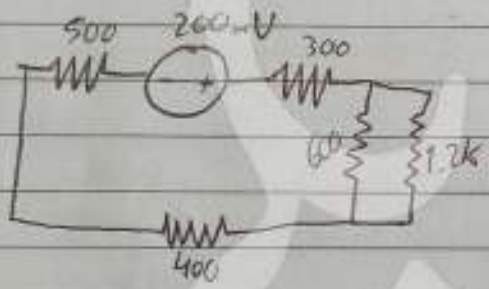
3 is in series with 5 and 7 → 3+5+7 = 15Ω

10 is parallel to 15 → 6Ω

4 is in series with 6 → 6+4 = 10Ω



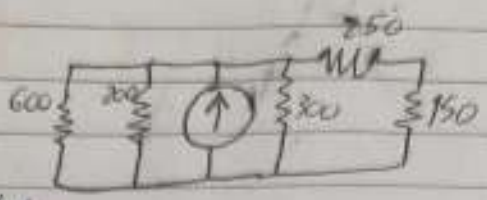
Q



Find Req

Solution:

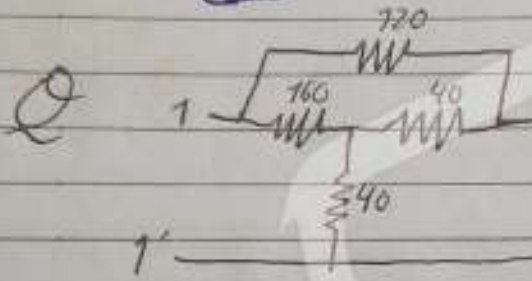


Q 

Find Req

Solution:

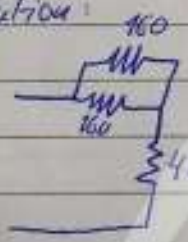


Q 

a) Find Req when 2-2' is O.C.

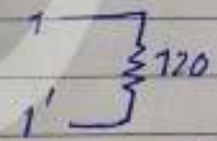
b) Find Req when 2-2' is S.C

Solution:

a) 

160 is parrallel to 160 → 80

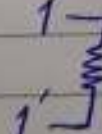
40 is in series with 80 → 40+80=120



b) 40 is parrallel to 40 → 20

160 is in series with 20 → 160+20=180

120 is parrallel to 180 → 72



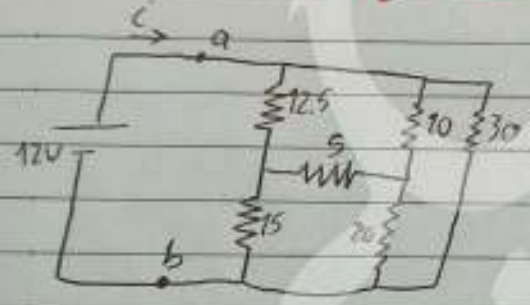
Δ ↔ Y Transformation



$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$
$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$
$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$



$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$
$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$
$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

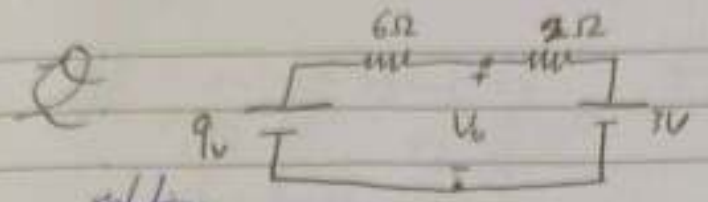




$$R_1 = \frac{15 \times 5}{40} = 1.875$$

$$R_2 = \frac{20 \times 5}{40} = 2.5$$

$$R_3 = \frac{20 \times 15}{40} = 7.5$$



Find V_o.

solution:

$-9 + 6i + 2i + 3 = 0$

$i = \frac{6}{8} A$

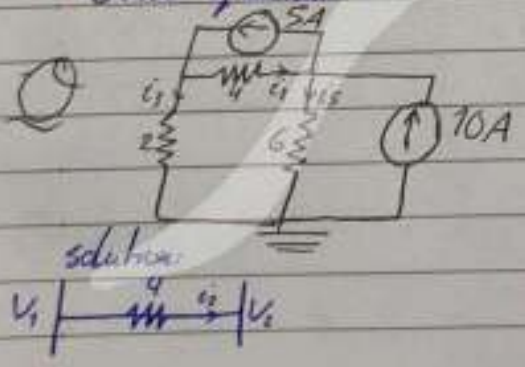
$-3 - 2(\frac{6}{8}) + V_o = 0$

$V_o = 4.5\ v$

Method of analysis

Nodal Method:

- To select reference node:  $V_{ref} = 0$
- To assign node voltages: $V_1, V_2, \ldots, V_{n-1}$
- Apply KCL to each node.
- Solve equations



Find i_1, i_3, i_4.

solution:

① $5 = i_3 + i_4$

$= \frac{V_1 - 0}{2} + \frac{V_1 - V_2}{4} \Rightarrow 20 = 3V_1 - V_2$

② $10 + \frac{V_1 - V_2}{4} = 5 + \frac{V_2 - 0}{6} \Rightarrow 60 = -3V_1 + 5V_2$

$20 = 3V_1 - V_2$

$60 = -3V_1 + 5V_2$

$80 = 4V_2 \Rightarrow V_2 = 20\ v$

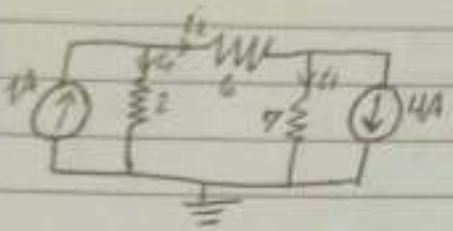
$V_1 = 13.33\ v$

$i_3 = \frac{V_1}{2} = 6.665 A$

$i_2 = \frac{V_2 - V_1}{4} = -1.6675$

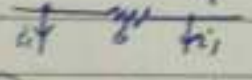
$i_4 = \frac{V_2}{6} = 3.33 A$

Q



Find i_1, i_2, i_3

Solution:



① $1 = \frac{V_1}{2} + \frac{V_1 - V_2}{6} \Rightarrow 6 = 4V_1 - V_2$

② $\frac{V_1 - V_2}{6} = \frac{V_2}{7} + 4 \Rightarrow 168 = 7V_1 - 13V_2$

$$\begin{bmatrix} 4 & -1 \\ 7 & -13 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 168 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 4 & -1 \\ 7 & -13 \end{vmatrix} = 4(-13) - (7 \cdot (-1)) = -45$$

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{\begin{vmatrix} 6 & -1 \\ 168 & -13 \end{vmatrix}}{-45} = -2V$$

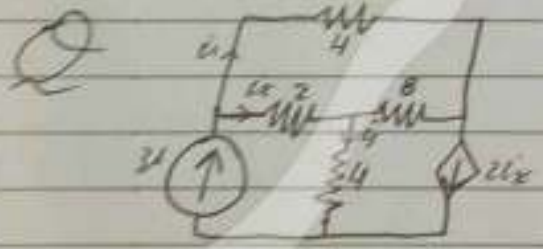
$$V_2 = \frac{\Delta_2}{\Delta} = \frac{\begin{vmatrix} 4 & 6 \\ 7 & 168 \end{vmatrix}}{-45} = -14V$$

$$i_1 = \frac{V_1}{2} = -1A$$

$$i_2 = \frac{V_1 - V_2}{6} = 2A$$

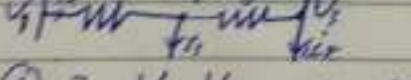
$$i_3 = \frac{V_1}{7} = -2A$$

Q



Find i_1, i_2, i_3

Solution



① $3 = \frac{V_1 - V_2}{4} + \frac{V_3 - V_2}{2}$

② $\frac{V_1 - V_2}{2} = \frac{V_2}{4} + \frac{V_2 - V_3}{8}$

③ $2i_x = \frac{V_2 - V_3}{8} + \frac{V_1 - V_3}{4}$

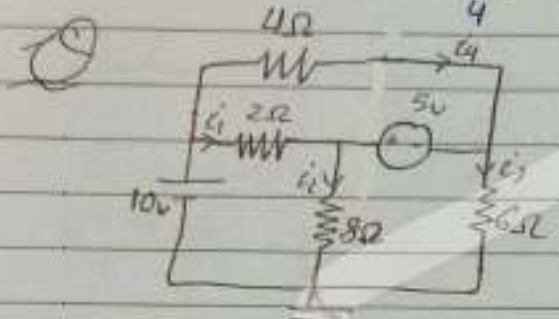
$$3V_1 - 2V_2 - V_3 = 12$$
$$-4V_1 + 7V_2 - V_3 = 0$$
$$24 - 3V_2 + V_3 = 0$$

$$\begin{bmatrix} 3 & -2 & -1 \\ -4 & 7 & -1 \\ 2 & -3 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}$$

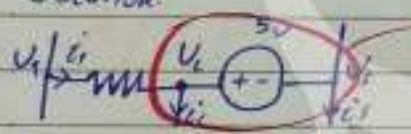
$$\Delta = \begin{bmatrix} 3 & -2 & -1 \\ -4 & 7 & -1 \\ 2 & -3 & 1 \\ 3 & -2 & -1 \\ -4 & 7 & -1 \end{bmatrix} = 3(7)(1) + (-4)(-3)(-1) + (2)(-2)(-1) + (-1)(7)(2) + (-1)(-3)(3) + (1)(-2)(-4) = 10$$

$V_1 = 4.8V$
$V_2 = 2.4V$
$V_3 = -2.4V$

$i_1 = \frac{V_1 - V_4}{4} = 1.8\,A$
$i_2 = \frac{V_2 - V_3}{8} = 0.6A$
$i_3 = \frac{V_4}{4} = 0.6A$



Solution:



Super node ⇒



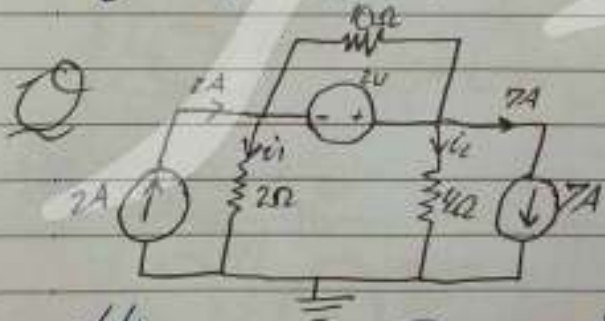
$i_1 + i_4 = i_2 + i_3$

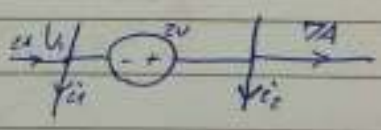
$\frac{V_1 - V_2}{2} + \frac{V_1 - 5}{4} = \frac{V_2}{8} + \frac{V_2}{6}$

$+5V-$

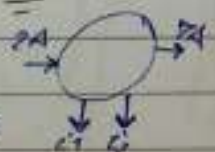
$+ V_2 -$ $\quad$ $+ V_3 -$

$-V_2 + 5 + V_3 = 0$





solution



$2 = i_1 + i_2 + 7$

$2 = \frac{V_1}{2} + \frac{V_2}{8} + 7$

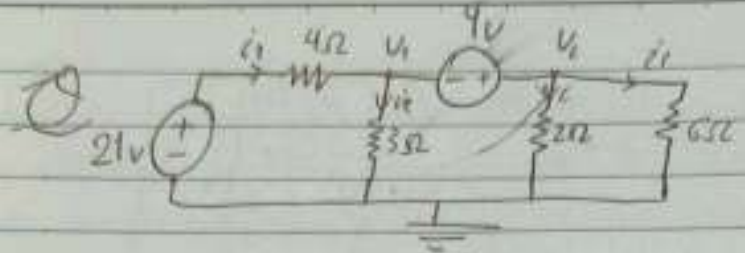
$V_1 = -7.33\,V$
$V_2 = -5.33\,V$

$-2V+$

$+ V_1 -$ $\quad$ $+ V_2 -$

$-V_1 - 2V + V_2 = 0$

$V_2 - V_1 = 2$

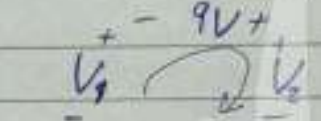


Solution:



$i_1 = i_2 + i + i_3$

$\frac{21-4}{4} = \frac{V_1}{3} + \frac{V_2}{2} + \frac{V_2}{6}$



$-V_1 - 9 + V_2 = 0$

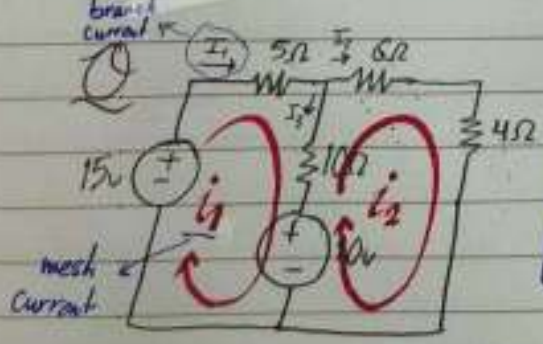
$V_2 - V_1 = 9$

Mesh Current Method



Loop: هي حلقة مغلقة

mesh: هي حلقة مغلقة لا تحتوي على حلقة صغيرة



Solution

① $-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$

② $-10 + 10(i_2 - i_1) + i_2(6) + i_2(4) = 0$

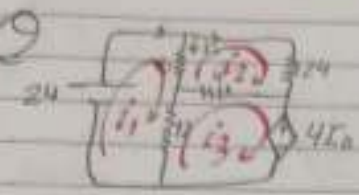
$3i_1 - 2i_2 = 1$

$2i_2 - i_1 = 1$

$i_1 = 1A$

$i_2 = 1A$

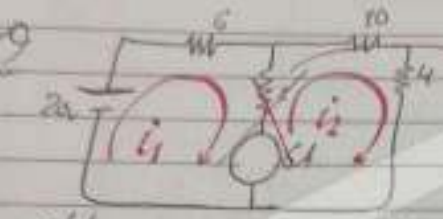
Q



Solution

$I_o = i_1 - i_2 - i_3 \Rightarrow$ لا يوجد

Q



→ i here equals zero

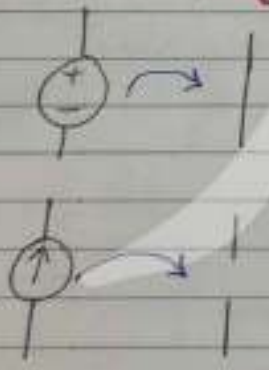
Solution:



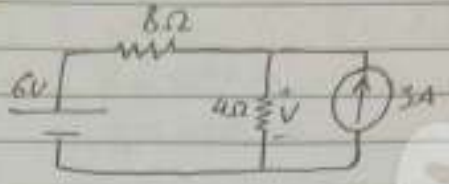
$-20 + 6i_1 + 10i_2 + 4i_2 = 0$

Circuit Theorems

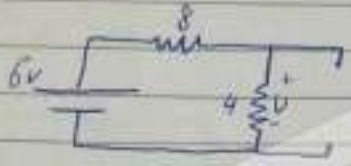
Superposition when having more than a source we start by calculating one by one and deleting other sources



Q Find V



Solution



$V = \frac{6 \times 4}{12} = 2v$

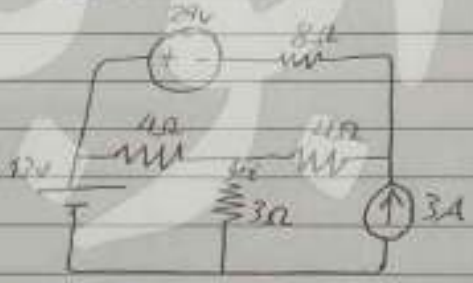


$I_{4\Omega} = \frac{8}{12} = 2A$

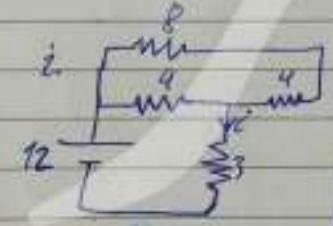
$V_{4\Omega} = 2 \times 4 = 8v$

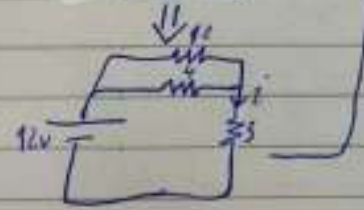
$V_{3A} = 8 + 2 = 10v$

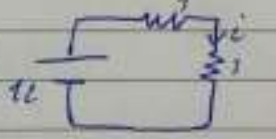
Q Find i.



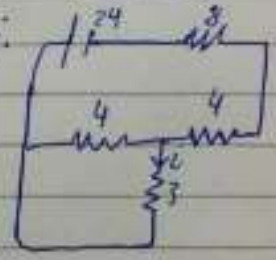
Solution







$\frac{12}{6} = \alpha Ai$

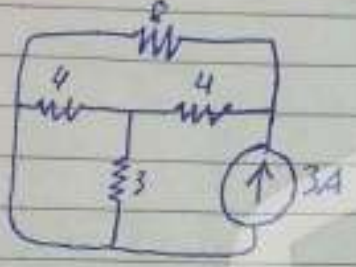




$24 + 8i_1 + 4i_1 + 4(i_1 - i_2) = 0$ ... ①

$4(i_2 - i_1) + 3i_2 = 0$ ... ②

$i_1 = -1A = i$



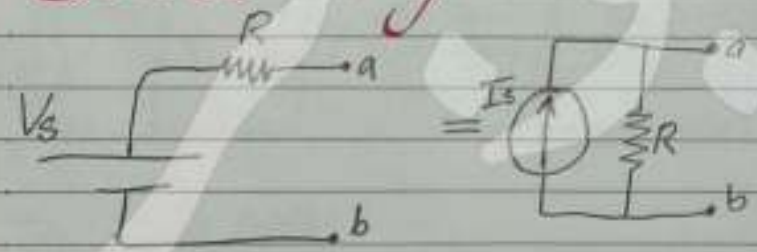
$\frac{V_2 - V_1}{4} = \frac{V_1}{3} + \frac{V_1}{4}$ ... ①

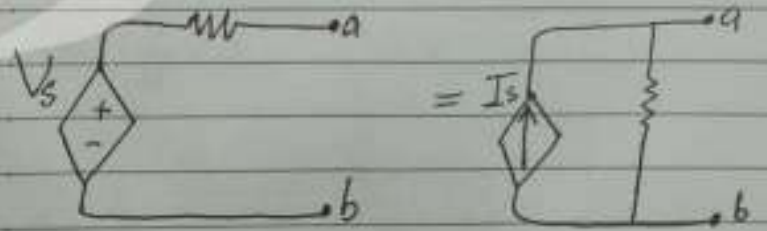
$3 = \frac{V_2 - V_1}{4} + \frac{V_1}{8}$ ... ②

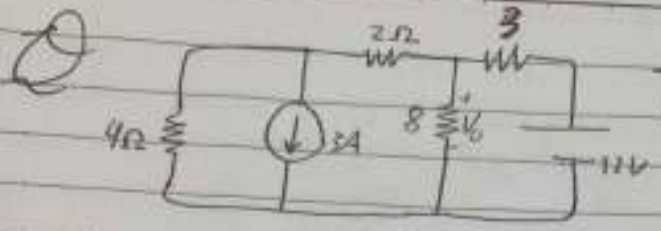
$i = \frac{V_1}{3} = 1A$

$i = 2 + -1 + 1 = 2A$

Source Transformation

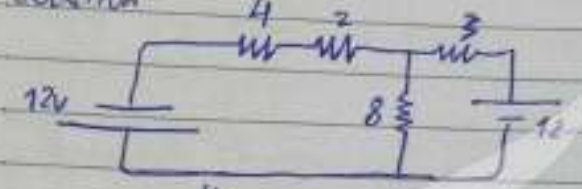


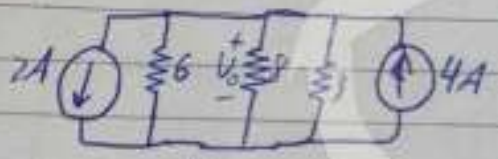




Find V_o.

Solution



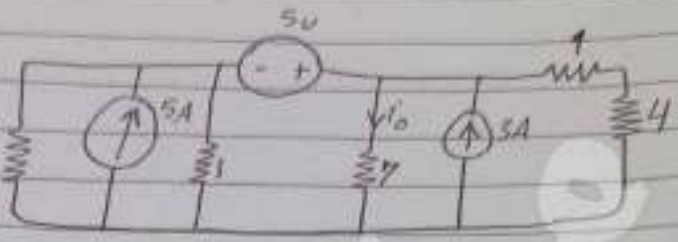




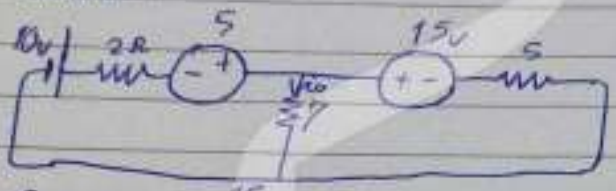
$$R_{eq} = \frac{1 \times 2}{10} = 0.4\Omega$$

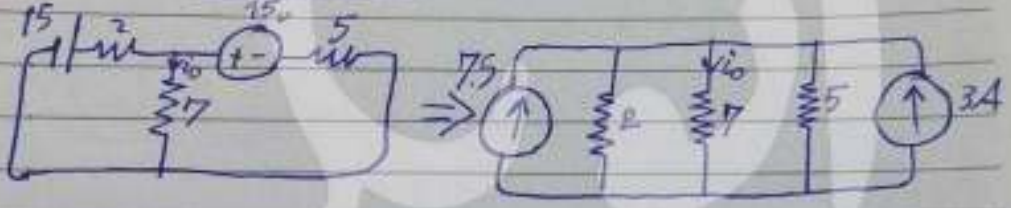
$$V_o = \frac{4}{10} \times 8 = 3.2\ V$$

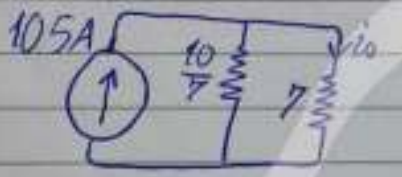
Q Find i_o.



solution:

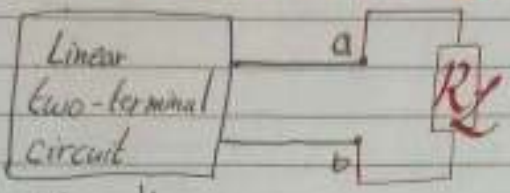




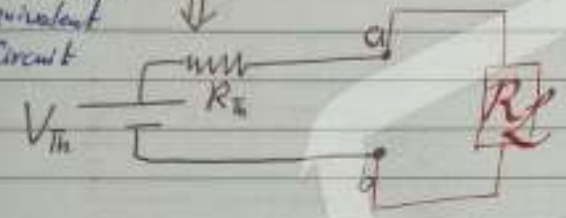


$$i_o = \frac{10.5 \times \frac{10}{7}}{\frac{10}{7} + 7} = 1.78A$$

Thevin's & Norton's Theorems

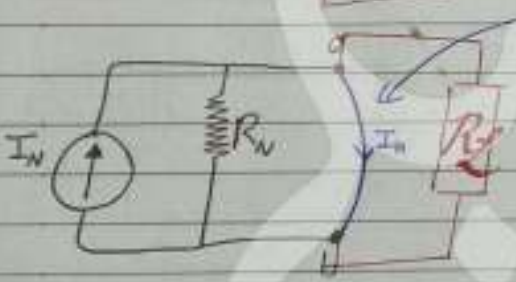


Thevinin's equivalent Circuit



$V_{Th} = V_{oc}$ at terminals a→b

$R_{Th} = R_{eq}$ | All independent sources are turned off



$I_N = I_{s.c}$ at a→b

$R_N = R_{eq} = R_{Th}$ | All independent sources are turned off

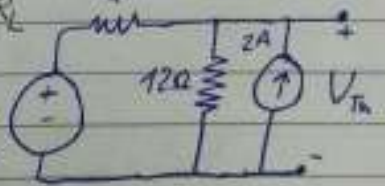
$I_N = \frac{V_{Th}}{R_{Th}}$

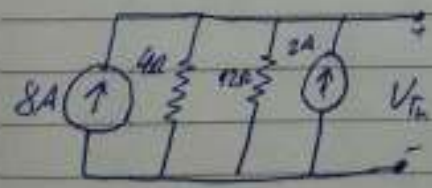
Q



Find V_{Th}:

Solution



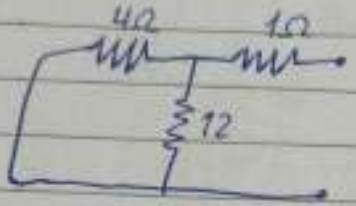




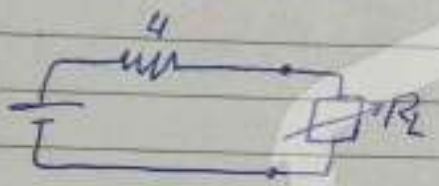
$I_{12\Omega} = \frac{10 \times 4}{16} = \frac{10}{4}\,A$

$V_{12\Omega} = V_{OC} = \frac{10}{4} \times 12 = \frac{120}{4} = 30V$

R_{Th}



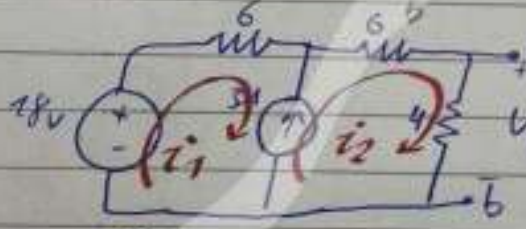
$R_{Th} = 1 + \frac{4 \times 12}{16} = 4\Omega$

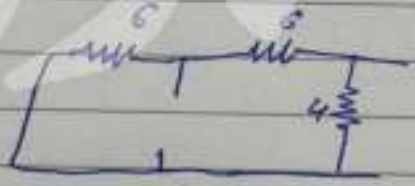




Find $I_{1\Omega}$.

Solution:





$R_{Th} = \frac{12 \times 4}{16} = 3\Omega$

$-18 + 6i_1 + 6i_2 + 4i_2 = 0$

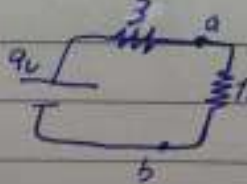
$i_2 = 3 + i_1$

$i_1 = -\frac{3}{4}\,A$

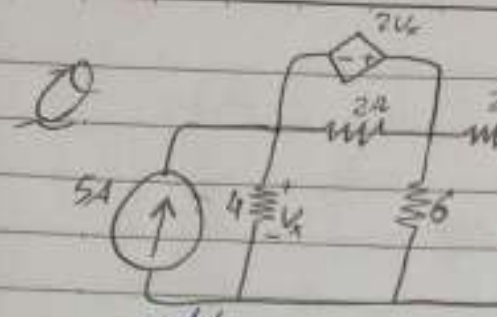
$i_2 = \frac{9}{4}\,A$

$V_{Th} = V_{OC} = i_2(4)$

$V_{Th} = \frac{9}{4} \times 4 = 9V$

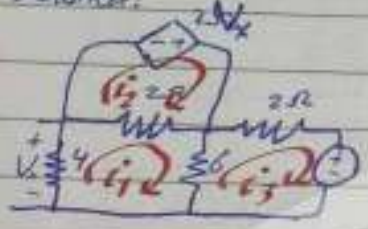


$I_{1\Omega} = \frac{9}{4}A$



اضافه کن!

Solution:



$i_3 = -i_o$

$V_x = 4i_1$

$V_x = i_1 - i_2$

$4i_2 + 2(i_2 - i_3) + 6(i_2 - i_3) = 0$

$6(i_3 - i_2) + 2i_2 + 1 = 0$

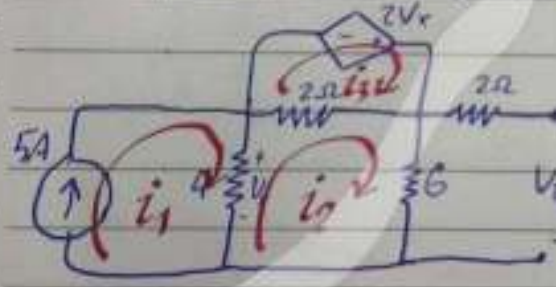
$R_{Th} = \frac{V_o}{i_o}$

To assume $V_o = 1v$

$R_{Th} = \frac{1}{i_o}$

$*i_o = -3i_2$

$R_{Th} = 6\Omega$



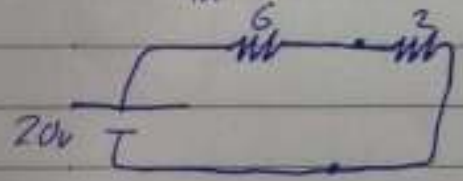
$i_1 = 5A$

$-2V_x + 2(i_3 - i_2) = 0$

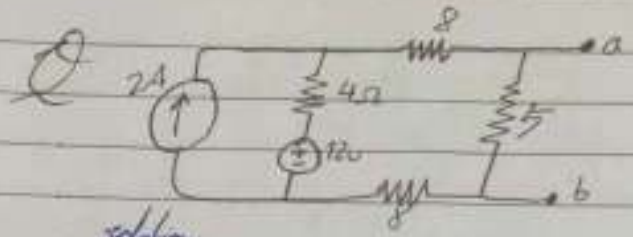
$12i_2 - 2i_3 - 20 = 0$

$i_o = \frac{10}{3}A$

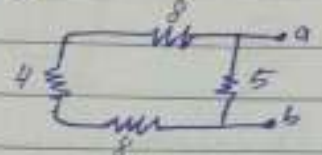
$V_{Th} = 20v$



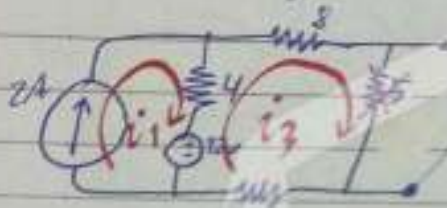
Q



solution



$R_N = 5 \| 20 = \frac{5 \times 20}{25} = 4\Omega$



$i_1 = 2A$

$-12 - 4i_2 + 5i_2 = 0$

$i_2 = 4A$

$i_2 = i_{S.C} = I_N = 1A$



Maximum Power Transfer



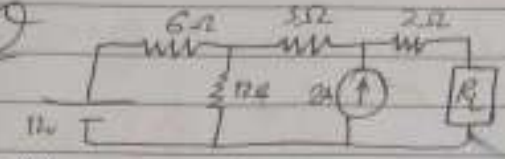
$P = i^2 R_L$

$P_{max} = \left(\frac{V_{Th}}{R_L + R_{Th}}\right)^2 R_L \Big|_{R_L = R_{Th}}$

$= \frac{V_{Th}^2}{4 R_{Th}^2} R_{Th}$

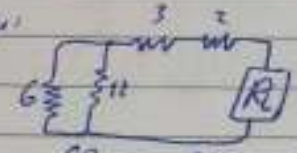
$P_{max} = \frac{V_{Th}^2}{4 R_{Th}} \Big|_{KL = R_{Th}}$

Q

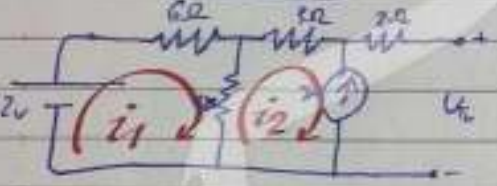


Find P_{max} at R_L

solution:



$R_L = R_{eq} = 9\Omega$



$i_2 = -2A$

$-12 + 6i_1 + 12i_1 - 12i_2 = 0$

$i_1 = -\frac{2}{3}A$

$-12 + 6i + 3(i_1) + V_{th} = 0$

$V_{Th} = 22v$

$P_{max} = \frac{22^2}{4 \times 9} = 13.44$ watt

Capacitors & Inductors

- passive elements
- reactive elements

C — stores

L

R — pure passive

Capacitance

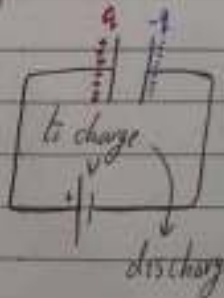
$C = \frac{\varepsilon A}{d}$ → area, d → distance

ε → constant of dielectric

↳ free space 8.85×10^{-12} F/m

F (μF, nF, pF)

air paper



$\Rightarrow CV = q$

$i = \frac{dq}{dt}$

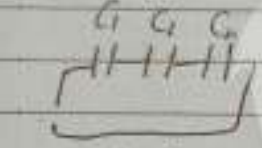
$i_c = C \frac{dv}{dt}$

$V = \frac{1}{C} \int_{t_0}^{t} i\,dt + V(t_0)$

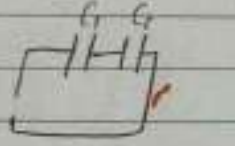


$i = C \frac{dv}{dt}$ → constant

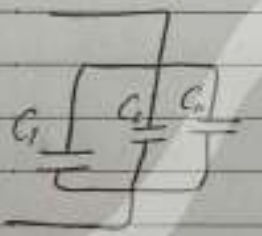
$\frac{dv}{dt} = 0 \Rightarrow i = 0 =$ ∞



$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}$



$C_T = \frac{C_1 C_2}{C_1 + C_2}$



$C_T = C_1 + C_2 + \dots + C_n$

$P = vi$

$P = C v \frac{dv}{dt}$

$W = \int_J P\,dt \rightarrow W_c = \frac{1}{2} C v^2$

Q $V_c = 10\cos 6000t$ V

$C = 5\mu F$, Find i?

Solution:

$$i = C\frac{dv}{dt}$$

$$= 5\times10^{-6}\frac{d(10\cos 6000t)}{dt}$$

$$= -5\times10^{-6}\times10\times6000\times\sin(6000t)$$

$$i = -.03\sin 6000t \text{ A}$$

Q Find V_c if $C = 2\mu F$

$i(t) = 6e^{-1000t}$ mA

$v(0) = 0$

Solution:

$$V_c = \frac{1}{C}\int_0^t 6e^{-1000t}\,dt + 0$$

$$= \frac{1}{2\times10^{-6}}\times\frac{6\times10^{-3}}{-1000}\left(e^{-1000t}\Big|_0^t\right)$$

$$V_c = 1 - e^{-1000t} \text{ V}$$

Q The current through a $100\mu F$ capacitor is $i(t) = 50\sin 120\pi t$ mA. Calculate the voltage across it at $t = 1$ ms.

Solution

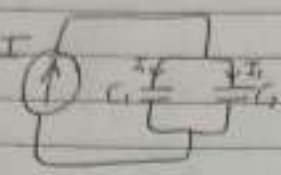
$$V_c = \frac{1}{C}\int_0^t 50\sin 120\pi t\,dt \times 10^{-3}$$

$$= \frac{1}{100\times10^{-6}}\cdot\frac{50\times10^{-3}}{120\left(\frac{22}{7}\right)}\left(\cos 120\pi t\Big|_0^{1\times10^{-3}}\right) = 93 \text{ mV}$$



$$V_1 = \frac{VC_2}{C_1+C_2}$$

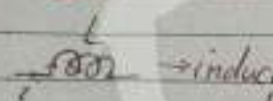
$$V_2 = \frac{VC_1}{C_1+C_2}$$



$$I_1 = \frac{I\,C_1}{C_1+C_2}$$

$$I_2 = \frac{I\,C_1}{C_1+C_2}$$

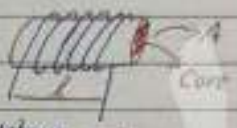
Inductor



→ inductance

$$L = \frac{N^2 \mu A}{l}$$

(H)



$$V_L = L\frac{di}{dt}$$



$$V_L = L\frac{di}{dt} = 0$$

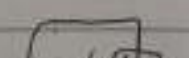
$$i = \frac{1}{L}\int_{t_0}^{t} v(t)\,dt + i(t_0)$$

$$V_L = 0 \rightarrow \text{sm} = \text{S.C}$$

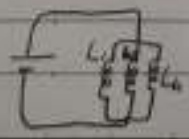
$$P = V_L = L_1 i \frac{di}{dt}$$

$$W = \int P\,dt$$

$$W = \frac{1}{2} L i^2$$

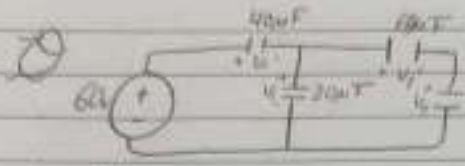


$$L_{total} = L_1 + L_2 + \ldots + L_n$$



$$L_{total} = \frac{1}{L_1} + \frac{1}{L_2} + \ldots + \frac{1}{L_n}$$

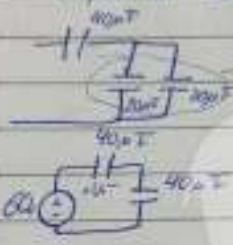
Q



Find V_1, V_2, V_3 and V_4

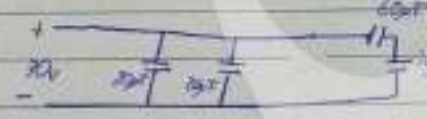
Solution

Capacitor 60μF is in series with 30μF. $C_T = \frac{60 \times 30}{90} = 20\mu F$



$C_T = 20 + 20 = 40\mu F$

$V_1 = 30V$



$V_1 = 30V$

$V_2 = \frac{30 \times 30}{90} = 10V$

$V_4 = 20V$

check $V_1 + V_2 + V_4 = 60V$

$V_1 // V_2 \cdot V_4$

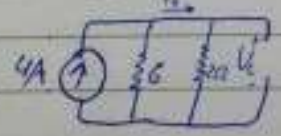
Q



Find V_c, i_L, W_c, W_L

Solution

In Dc domain



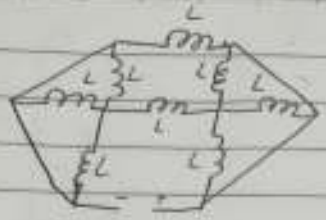
$i_L = i_{sc} = \frac{4(6)}{8} = 3A$

$V_{oc} = V_{oc} = V_c = 3 \times 2 = 6V$

$W_c = \frac{1}{2} C V^2 = \frac{1}{2} \cdot 4 \times 36 = 72J$

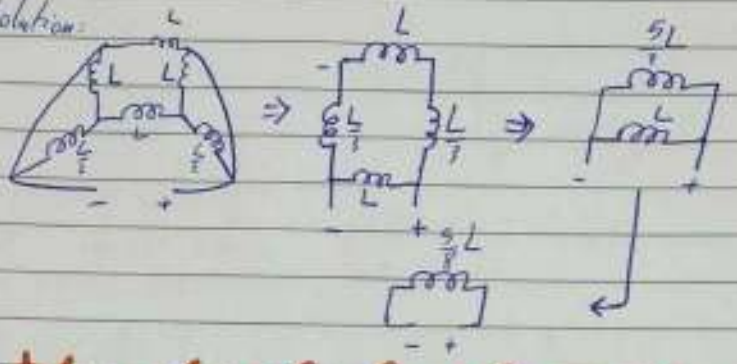
$W_L = \frac{1}{2} L i^2 = \frac{1}{2} \cdot 6 \cdot 9 = 27J$

Q



Find L_{eq}

Solution:



First Order Circuits

i. Source Free Circuits → RL, RC

Independent Source Circuit

Assume:

$V_c(t) = V_c(t=0) = V_0$

$W_c(t=0) = \frac{1}{2} C V_0^2$



Using KCL:

$i_R + i_C = 0$

$C\frac{dv}{dt} + \frac{V}{R} = 0$

$\frac{dv}{dt} = -\frac{V}{RC}$

$\int \frac{dv}{V} = -\int \frac{dt}{RC}$

$\ln V = -\frac{t}{RC} + \ln A$

$\ln V - \ln A = -\frac{t}{RC}$

$\ln \frac{V}{A} = -\frac{t}{RC}$

$\frac{V}{A} = e^{-\frac{t}{RC}}$

$V(t) = Ae^{-\frac{t}{RC}}$

Initial condition $t=0 \rightarrow V_c(t) = V_0$

$V_0 = Ae^0$

$A = V_0$

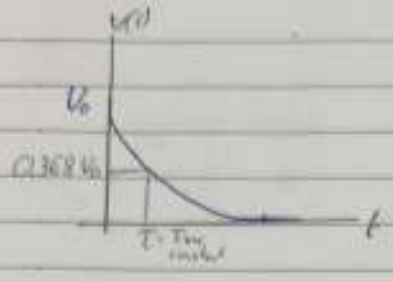
$V(t) = V_0 e^{-\frac{t}{RC}}$

$RC = \tau$: time constant

$V(t) = V_0 e^{-\frac{t}{\tau}}$

Natural Response

exponential Decay



$t = 5\tau$

$V(t) = V_0 e^{-\frac{5\tau}{\tau}}$

$= \frac{10 \times 0.00673}{0.00673}$

5τ

Capacitor is is fully discharged



$V_c(0) = 15V$ at $t > 0$

Find V_c, V_x, i_x.

solution:

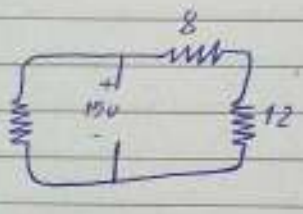
$V(t) = V_0 e^{-\frac{t}{\tau}}$

$R_{eq} = 4\Omega$

$\tau = R_{eq} C = 0.4s$

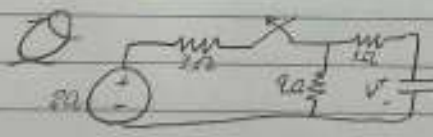
$V_c = 15e^{-\frac{t}{0.4}}$

$V_c = 15e^{-2.5t}$ V



$V_x = 15e^{-2.5t}\left(\frac{12}{12+3}\right) = 9e^{-2.5t}$ V

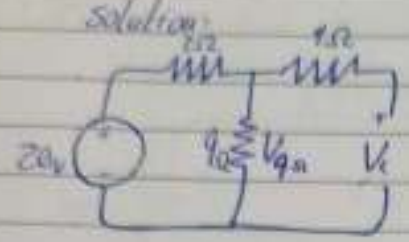
$i_x = \frac{V_x}{12} = 0.75e^{-2.5t}$ A



The switch in the following circuit has been closed for a long time & it is opened at $t = 0$

Find $v(t)$ for $t \geq 0$ calculate initial energy stored in the capacitor.

Solution:



$V_{1\Omega} = V_c = \frac{20 \times 9}{12} = 15V$

$V_c(0^-) = V_c(0) = 15V$

$V(t) = V_0 e^{-\frac{t}{\tau}}$

$= 15e^{-\frac{t}{\tau}}$

$\tau = RC$

$= 10(20 \times 10^{-3})$

$= 200ms = 0.2s$

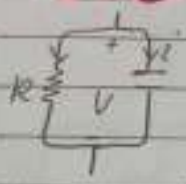
$V(t) = 15e^{-\frac{t}{0.2}}$

$V(t) = 15e^{-5t}$

$W(0) = \frac{1}{2}CV_0^2 = \frac{1}{2} \times 20 \times 10^{-3}(15)^2 J$



RC

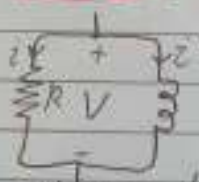


$V(t) = V_0 e^{-\frac{t}{\tau}}$

$\tau = RC$

$V(0^-) = V(0)$

RL



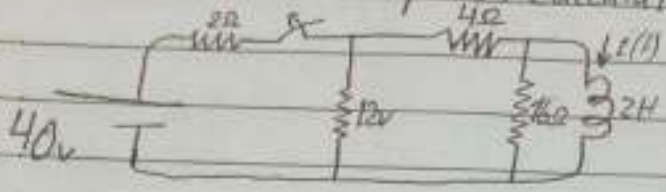
$i(t) = I_0 e^{-\frac{t}{\tau}}$

$\tau = \frac{L}{R}$

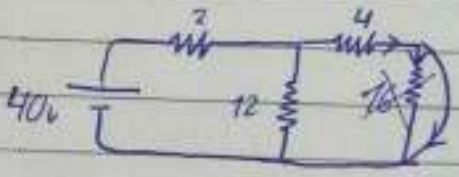
$i(0^-) = i(0)$

نافع وجميع العلم النافع فيها الخلق

Q. The switch has been closed for long time. At t=0, then the switch is opened. Calculate $i(t)$ for $t>0$.

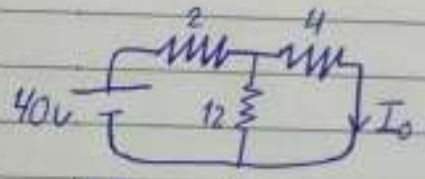


solution:

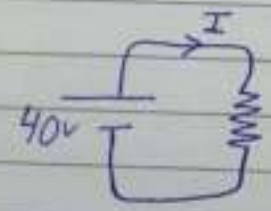


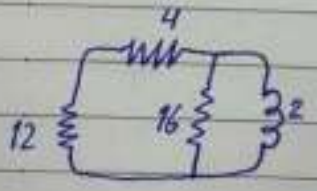
$$R_{eq} = \frac{4\times12}{16} + 2 = 5\Omega$$

$$I = \frac{40}{5} = 8A$$

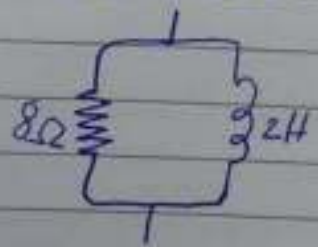


$$I_o = 8\times\frac{12}{16} = 6A$$





RL-Circuit



$$\tau = \frac{L}{R} = \frac{1}{4}$$

$$i(t) = 6e^{-\frac{t}{\tau}}$$

$$i(t) = 6e^{-\frac{t}{\frac{1}{4}}}$$

$$i(t) = 6e^{-4t}\ A$$